CESIM TEACHING NOTE

Human Resource Planning Microsimulation

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1. Simulation Overview

The Human Resource Planning (HRP) simulation is a **single-player** business simulation that places participants in the role of HR decision-makers within a fast-growing company. It offers a realistic, interactive environment where students apply human resource strategies to address evolving business needs. The simulation emphasizes core areas such as demand and supply forecasting, talent acquisition, compensation planning, training, and HR policy implementation.

Learning Objectives

Upon completion of this simulation, participants will be able to:

- **Strategic HR Alignment**: Understand how to align human resource planning with overall business objectives to support organizational growth and effectiveness.
- **Workforce Forecasting**: Develop skills in accurately predicting human resource demand and supply to ensure the right talent is available at the right time.
- **Effective recruitment Strategy**: Understand how to select effective hiring channels and establish a strong employer brand to attract quality talent.
- **Compensation and benefit planning**: Make data-driven decisions about salary structures and benefits that balance cost-efficiency with employee satisfaction and retention.
- **Training & Development:** Identify and prioritize learning initiatives that close skill gaps, promote employee growth for future organizational needs.
- **HR Policy Design & Implementation:** Create and implement HR policies that enhance employee well-being, support compliance, and contribute to overall organizational performance.

Target Audience

- **Undergraduate students** in human resource management, organizational behaviour, and business administration courses
- **Graduate students** in MBA programs, MS in HR Management, or related fields
- **Executive education participants** seeking to enhance their strategic HR planning capabilities
- **Corporate training participants** in HR departments looking to develop workforce planning skills
- **HR professionals** interested in strengthening their analytical and strategic decision-making abilities

Time Required

- 60-90 minutes for individual simulation completion
- 45-60 minutes for guided post-simulation debrief

Case Scenario and Winning Criteria

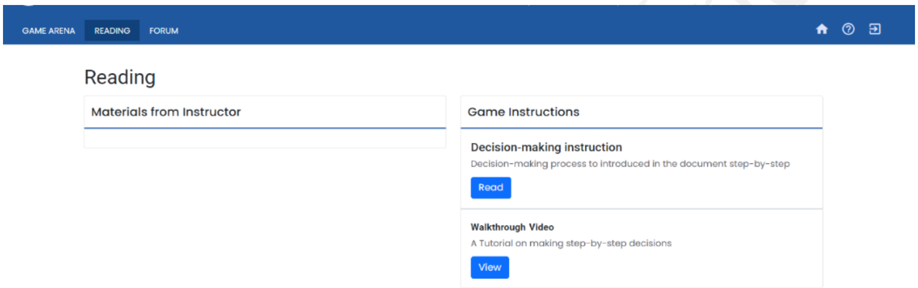
In the Cesim HRP simulation, participants take on the role of HR managers in a fast-growing organization operating in a competitive environment. Their task is to align human resource strategies with overall business objectives, make informed workforce decisions, and respond to changing conditions that impact organizational performance. The ultimate objective is to **achieve employee satisfaction score as >8.5/9.5** to enhance retention, engagement, and productivity during expansion

Simulation Structure

The microsimulation is organized into independent rounds, each presenting either consistent scenarios or allowing the instructor to customize market conditions to simulate decision-making challenges throughout the company's growth journey.

2. Pre- Simulation Preparation

a. **Review Background Materials**



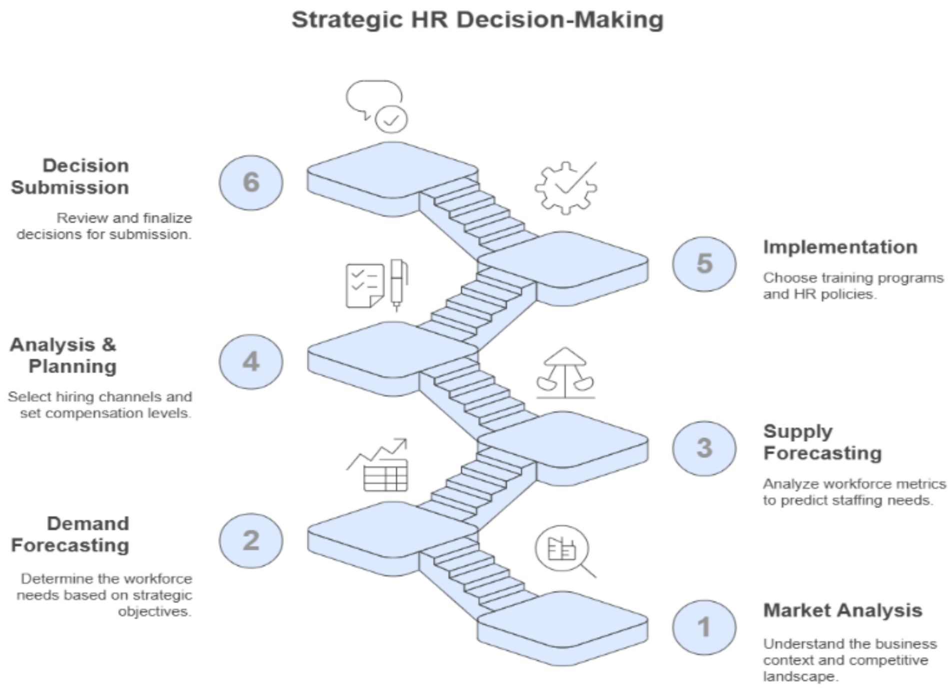
- Login as participant to explore the platform
- Read the decision-making guide thoroughly
- Watch the walkthrough video for practical insights
- Understand the company context and business objectives

b. **Familiarize with the Interface**

- Explore the Game Arena layout
- Understand the sequence of decision tabs
- Review the "Food for Thought" section for critical considerations

Understanding Decision-Making Process

The simulation follows a logical step-by-step process. Participants should move through each stage to make thoughtful and informed HR decisions.



The simulation evaluates performance based on three core algorithms:

1. **Workforce Alignment Algorithm**: Evaluates how well staffing decisions match business requirements across departments. Strong alignment between workforce planning and strategic initiatives produces higher organizational efficiency.
2. **Talent Acquisition Efficiency Model**: Calculates time-to-fill and cost-per-hire based on the match between recruitment channels and position types. Different roles respond differently to various recruitment approaches.
3. **Employee Experience Impact System**: Tracks how compensation, training, and policy decisions affect satisfaction and retention metrics based on departmental and demographic factors.

Key Instructor-Only Parameters

- **Departmental Elasticity Factors**: The backend shows exact response rates of different departments to compensation changes. Example: Technical departments show 0.85 elasticity (15% retention improvement per 10% compensation increase) versus 0.65 for administrative roles.
- **Recruitment Channel Effectiveness Multipliers**: Campus recruitment yields 3.2x more qualified technical candidates than job portals, but with 2.5x longer time-to-fill.
- **Training Impact Coefficients**: Technical training produces 18% efficiency gains in technical departments but only 5% in administrative functions.
- **Policy Satisfaction Multipliers**: Flexible work policies increase satisfaction by 0.8 points in technical roles but only 0.3 points in customer-facing positions.
- **Attrition Risk Thresholds**: Departments with satisfaction scores below 6.8 experience exponential increases in attrition rates (2x base rate), while scores above 8.2 reduce attrition by 40%.

Teaching with This Information

- During debriefing, use these parameters to explain why certain strategies succeeded or failed. This transforms the simulation from a "black box" into a transparent demonstration of strategic HR principles.
- Reveal how seemingly minor decisions (like a 2% difference in compensation strategy) can produce significant outcome differences due to the elasticity factors and threshold effects built into the model.
- Use the backend data to challenge common HR misconceptions, such as "all departments respond equally to the same interventions" or "more recruitment channels always yield better results."

Common Participant Challenges

- **Analysis Paralysis**: Guide participants to focus on key workforce metrics rather than trying to analyze all available data.
- **Inconsistent Strategy**: Point out contradictory decisions (e.g., aggressive internal promotion targets without corresponding training investments).
- **Departmental Homogenization**: Use the simulation data to challenge assumptions about uniform approaches across different business functions.
- **Overlooking Strategic Alignment**: Emphasize how decisions in HR planning, recruitment, compensation, and development must reinforce each other to create coherent strategy.
- **Cost-Minimization Bias**: Help students recognize when short-term cost reduction undermines long-term organizational capability.

Instructor Pre-Implementation Checklist for Simulation Delivery

To ensure a smooth and impactful simulation experience for your learners, please review and complete the following preparatory steps:

- Familiarize Yourself with the Simulation: Complete a test run of the simulation to understand its structure, decision flow, and pacing. This allows you to better anticipate participant questions and challenges.
- Review Simulation Mechanics: Examine the instructor-only backend settings and key variables (e.g., cost structures, demand drivers, elasticity assumptions) to understand how decisions influence outcomes.
- Align with Course Objectives: Identify the core concepts from your course that align with the simulation to reinforce key learning goals.
- Prepare Contextual Examples: Bring 2–3 real-world examples from relevant industries to illustrate successful and unsuccessful strategic decisions during the debrief.
- Review Instructor Insights and Scoring: Familiarize yourself with the instructor account, leaderboard and performance indicators available to instructors. These tools will help guide feedback and encourage reflective learning.
- Ensure Participant Access; Confirm that all participants have registered using the right link / code and are able to access the simulation platform before the session begins.
- Test Technical Readiness: Coordinate with your IT team (if applicable) to ensure all devices, software, and internet connectivity are ready and compatible with the simulation requirements.

3. Facilitation Roadmap

Pre-Simulation Briefing

Introduce the Business Context

- Present the company scenario and its strategic goal and journey
- Explain participant take the role of HR decision-makers supporting business growth
- Highlight the competitive talent landscape and strategic importance of workforce planning

Explain the Simulation Flow

- Walk through the decision sequence: market analysis → demand forecasting → supply planning → recruitment & compensation → training & policies
- Clarify the interconnections between decisions and how earlier choices constrain later options
- Demonstrate how to navigate between simulation tabs and interpret key data visualizations

Set Expectations

- Emphasize that decisions should be driven by business strategy and data analysis
- Explain that there is no single "correct" approach, but that strategic alignment is critical
- Prepare students for the need to balance competing priorities (cost, time, quality, satisfaction)

During Simulation

1. Business Context Phase

- Encourage thorough examination of company background and strategic objectives
- Suggest noting key growth targets, technology initiatives, and market conditions
- Prompt participants to identify which departments will be most critical to strategic success

2. Demand Forecasting Phase

- Guide students to analyze current staffing levels and departmental efficiency metrics
- Emphasize the importance of calculating additional workforce needs based on strategic initiatives
- Highlight the differential scaling requirements across departments during technology transformation

3. Supply Planning Phase

- Direct attention to current workforce dynamics (attrition, promotions, transfers)
- Encourage thoughtful consideration of internal mobility versus external hiring trade-offs
- Remind participants to account for cascading effects of promotions on vacancy creation

4. Recruitment & Compensation Phase

- Prompt comparison of different recruitment channels' effectiveness for various position types
- Guide analysis of market compensation data alongside internal equity considerations
- Suggest reviewing how compensation decisions might affect both attraction and retention

5. Implementation Phase

- Encourage selection of training programs that directly support strategic initiatives
- Direct students to consider the fit between HR policies and organizational culture
- Remind students to review the full set of decisions for coherence before final submission

Results Analysis

- Guide analysis of key performance metrics (satisfaction, efficiency, time-to-fill, cost)
- Prompt reflection on which decisions most significantly influenced outcomes
- Encourage identification of improvement opportunities for subsequent rounds

Debrief Structure

1. Performance Distribution

- Display anonymized class results showing the range of outcomes across key metrics
- Identify patterns distinguishing high-performing strategies from others
- Discuss the performance trade-offs evident in different approaches

2. Strategy Alignment Discussion

- Compare strategically aligned versus misaligned approaches from the class
- Show how decisions that reinforced business priorities led to superior outcomes
- Highlight examples where internal consistency across HR decisions strengthened results

3. Key Decision Analysis

- Review optimal approaches for critical decision areas (workforce planning, recruitment, development)
- Share the backend performance parameters explaining why certain choices performed better
- Discuss the threshold effects and elasticity factors that influenced outcomes

4. Theoretical Connections

- Link simulation experiences to key HR theories (strategic HRM, resource-based view, human capital theory)
- Connect simulation outcomes to workforce planning principles and talent management frameworks
- Demonstrate how simulation dynamics reflect real-world HR challenges and trade-offs

5. Real-World Applications

- Relate simulation insights to actual HR practices in technology and e-commerce companies
- Share examples of how organizations like Amazon, Google, or Infosys approach similar challenges
- Discuss how these strategic HR principles apply across different industries and organizational contexts

6. Action Learning Integration

- Guide participants to synthesize key learnings into practical takeaways
- Facilitate reflection on how simulation experiences might inform their future HR leadership
- Encourage application of insights to current workplace challenges (for executive participants)

4. Critical Decision Points Analysis

Decision Point 1: Workforce Demand Forecasting

Key Decision: Determining the additional workforce required to meet business objectives across different departments based on current staffing levels and efficiency metrics.

Optimal Approach: Success requires careful analysis of:

- **Current Workforce Assessment:** Evaluate staffing levels and efficiency across all departments to establish baseline capacity and identify over- or under-utilization.
- **Additional Hours Required:** Convert forecasted workload into full-time equivalents (FTEs), considering efficiency levels and departmental priorities.
- **Technology-Driven Growth Strategy:** Innovation-led initiatives may increase staffing needs by 30–40% in departments like Technical, Design & Production, and Customer Experience.
- **Revenue Expansion Plans:** Align Sales & Marketing staffing with projected growth in lead generation, campaigns, and customer acquisition.
- **Production Scalability:** Scale Design & Production teams to support higher output while balancing innovation with timely execution.
- **Logistics & Supply Chain Readiness:** Adjust logistics staffing to handle increased volume and maintain fulfillment and distribution performance.
- **Customer Experience Scaling:** Expand support teams to ensure service quality as customer volume increases across multiple channels.
- **Administrative Support:** Apply the 1:10 ratio—1 administrative hire for every 10 new hires across departments—for adequate operational support.
- **Departmental Interdependencies:** Account for cross-functional impacts and build flexibility to adapt to shifting business demands

Common Pitfalls:

- Underestimating technical staffing needs for technology implementation
- Failing to account for administrative support requirements (1:10 ratio)
- Applying uniform growth percentages across all departments regardless of strategic importance
- Overlooking efficiency improvements from technology adoption
- Disregarding current efficiency metrics when calculating new hiring needs
- Failing to recognize that insufficient hiring will result in contractual workers at 12% higher cost

Current Workforce Status & Additional Requirements

Department	Current Staff	Current Efficiency	Additional Hours Required
Technical	80	85%	31680
Sales & Marketing	60	90%	33440
Design & Production	40	87%	17600
Logistics & Supply Chain	40	80%	8800
Customer Experience	25	75%	12320
Administration	25	92%	Based on 1:10 ratio

Decision Point 2: Supply Gap Analysis & Internal Mobility

Key Decision: Determining how many positions to fill externally versus promoting internally, considering workforce metrics such as attrition, promotion rates, and internal transfers.
Optimal Approach: Data supports several viable internal mobility strategies:

- **Technical Career Pathway**: Prioritize internal promotions for 40-50% of technical positions to retain institutional knowledge, particularly critical during AR/AI implementation. This requires identifying ~8-9 internal candidates for upward mobility.
- **Leadership Development Strategy**: Promote 60-70% of management positions internally while recruiting fresh perspectives for specialized roles. This maintains cultural continuity while bringing in external expertise.
- **Balanced Pipeline Approach**: Maintain 30% internal promotion rate across all departments to balance retention and new talent acquisition. For the estimated 60+ new positions, approximately 18 should be filled through internal promotions.

Real Gap Calculation Example:
For Technical Department (requiring 18 new positions):

- Expected attrition: 80 × 8.2% = ~7 employees
- Total positions to fill = 18 + 7 = 25 positions
- With 40% internal promotion target = 10 internal moves, 15 external hires
- Each internal promotion creates another vacancy, potentially in another department

Common Pitfalls:

- Ignoring department-specific attrition rates when planning (Technical roles may have higher turnover)
- Creating career advancement bottlenecks by over-recruiting externally
- Promoting too aggressively without sufficient training support
- Failing to account for the cascading effect of internal promotions (each promotion creates another vacancy)
- Overlooking the time lag between identifying a vacancy and filling it (39 days average)
- Neglecting the relationship between employee satisfaction (7.68) and potential future attrition

Workforce Dynamics & Internal Mobility Metrics

Metric	Current Value	Impact on Workforce Planning
Attrition Rate	8.20%	Represents ~33 employees leaving annually
Promotion Rate	2%	Represents ~8 employees changing roles upward
Internal Transfer Rate	5%	Represents ~20 employees changing departments
Voluntary Resignation	2%	Subset of attrition that may be preventable
Employee Satisfaction	7.68/9.5	Indicator of retention potential
Time to Fill Position	39 days	Operational gap during hiring process

Decision Point 3: Recruitment Channel Selection

Key Decision: Selecting the most effective recruitment channels for different departments and position levels while strategically building employer brand.

Optimal Channel Mix Approach:

- **Technical Department**: Combine campus recruitment (40%) for junior roles, employee referrals (30%) for mid-level positions, and specialized agencies (30%) for senior technical roles leading AR/AI implementation.
- **Sales & Marketing**: Leverage social media (40%), online job portals (30%), and employee referrals (30%) to attract trend-conscious professionals who understand digital consumer behavior.
- **Design & Production**: Focus on specialized creative platforms, social media (50%), and employee referrals (30%) to attract design talent familiar with AR applications.
- **Administrative & Support Roles**: Primarily utilize cost-effective online job portals (70%) and referrals (30%).

Common Pitfalls:

- Relying on a single recruitment channel regardless of position type, limiting candidate diversity
- Overlooking the time-to-fill implications of channel selection, creating operational gaps
- Selecting high-cost recruitment options for easily fillable positions, driving up hiring costs
- Failing to balance immediate hiring needs with long-term pipeline development
- Neglecting to align recruitment channels with desired employer brand positioning
- Underestimating how channel selection affects the candidate experience and first impression of the company

Recruitment Channel	Cost per Hire (INR)	Candidate Quality	Best For
Online Job Portals	₹2,500	★★★☆☆	Entry to mid-level positions across departments
Campus Recruitment	₹4,000	★★★★☆ (for entry-level)	Technical roles, fresh talent pipeline, tech-savvy candidates
Employee Referrals	₹30,000	★★★★★	Mid-level positions requiring cultural alignment
Social media & Company Website	₹14,000	★★★☆☆	Creative roles, brand-conscious candidates
Recruitment Agencies & Head-hunters	₹60,000	★★★★☆ (for specialized roles)	Leadership positions, specialized technical roles

Decision Point 4: Compensation Strategy

Key Decision: Determining appropriate salary increases for new hires and existing employees while maintaining internal equity and market competitiveness.

Optimal Approach:

- **Market-Aligned Increases**: Offer 2-3% above market rate for critical skill areas (Technical, Design) where talent is essential to AR/AI implementation.
- **Internal Equity Balance**: Maintain new hire compensation within 5% of existing employee compensation for similar roles to prevent salary compression issues.
- **Retention-Focused Strategy**: Prioritize higher increases (8-10%) for high-performing employees in departments with higher attrition or strategic importance (Technical).
- **Differential Approach**: Implement targeted increases based on market scarcity (higher for Technical roles at 9-10%, moderate for Support functions at 6-7%).

Common Pitfalls and Their Repercussions:

- **Uniform Increases Across Departments**: Ignoring department-specific market dynamics leads to either overpayment in stable markets or critical talent loss in competitive areas.
- **Overlooking Market-Company Gap Trends**: Even departments currently above market (Sales & Marketing) face risk if the hike differential (1.2%) continues to widen year-over-year.
- **Focusing on Base Salary Without Considering Hike Percentages**: Employees often benchmark their increases against industry peers; consistently lower hikes signal limited growth potential even if base compensation is competitive.
- **Neglecting Internal Equity**: Creating significant compensation disparities between new and existing employees in the same roles leads to morale issues, knowledge hoarding, and increased turnover among experienced staff.
- **Short-term Cost Cutting**: Implementing below-market compensation to reduce immediate expenses typically results in higher long-term costs through increased turnover, extended vacancies, and reliance on contractors (at 12% premium).

Decision Point 5: Training & Development Investment

Key Decision: Selecting training programs that align with strategic initiatives and address skill gaps across departments while optimizing investment impact.

Optimal Approach:

- **Technology Transformation Support**: Prioritize AR training for Technical and Design teams (approx. 25-30 employees) to enable on-time technology implementation. This represents the critical path for strategic success.
- **Cross-Functional Alignment**: Implement coordinated training across interdependent departments to ensure seamless execution of new initiatives. E-commerce Analytics training should target both Technical and Sales departments simultaneously.
- **Targeted Skill Gap Remediation**: Address specific departmental weaknesses based on efficiency metrics. Customer Experience team (75% efficiency) would benefit significantly from Customer Service Excellence training.
- **New Hire Integration**: Include key training programs in onboarding plans for new employees, particularly in high-growth departments (Technical, Sales & Marketing), to accelerate productivity ramp-up.
- **ROI Optimization**: Balance high-cost strategic training (AR implementation) with high-ROI programs (Customer Service Excellence) to maximize overall development impact within budget constraints.

Common Pitfalls:

- **Disconnecting Training from Strategic Goals**: Implementing training programs without clear alignment to the AR/AI transformation and market expansion objectives.
- **Uniform Training Distribution**: Allocating training resources equally across departments rather than targeting strategic priorities and capability gaps.
- **Overlooking Participation Metrics**: Investing in programs without monitoring and addressing completion rates, which often fall below 50% without proper management support.
- **Focusing on Cost Without Strategic Value**: Selecting programs based primarily on cost rather than strategic impact, particularly problematic when AR/AI implementation is business critical.
- **Neglecting Post-Training Application**: Failing to create opportunities for employees to apply new skills immediately after training, reducing knowledge retention and practical value.

Training Program Analysis & Strategic Alignment

Training Program	Cost per Employee (INR)	Primary Target Departments	Efficiency Impact	Strategic Alignment
Digital Marketing Mastery	₹15,000	Sales & Marketing	+3%	★★★★☆ Essential for expanding market presence
Advanced E-commerce Analytics	₹20,000	Sales & Marketing, Technical	+3.5%	★★★★☆ Enables data-driven decision making
Augmented Reality (AR) in Fashion E-commerce	₹25,000	Technical, Design & Production	+4%	★★★★★ Critical for technology transformation
Customer Service Excellence	₹10,000	Customer Experience	+2%	★★★☆☆ Supports quality customer interactions
Sustainable Fashion and Ethical Production	₹18,000	Design & Production	+3.2%	★★★☆☆ Aligns with industry trends

Decision Point 6: HR Policy Implementation

Key Decision: Implementing HR policies that enhance employee satisfaction, retention, and organizational culture.

Optimal Approach:

- **Wellness-Centered Policies**: Highest impact on employee satisfaction (7-10% improvement) with moderate implementation costs
- **Learning & Development Policies**: Most effective for departments with higher proportions of millennials and Gen Z employees
- **Work Flexibility Policies**: Critical for retention in technical and creative roles
- **Inclusive Workplace Initiatives**: Essential for supporting diverse hiring and fostering innovation

Common Pitfalls:

- Implementing policies without considering their fit with company culture
- Selecting high-cost policies with marginal impact on key metrics
- Failing to match policies to the demographic profile of the workforce
- Overlooking the operational implications of policy implementation

HR Policy Analysis & Strategic Impact

Policy	Cost (INR)	Satisfaction Impact	Retention Impact	Departmental Relevance
Remote Work Flexibility Policy	₹500,000	+0.7 points	7% reduction in attrition	High Impact: Technical, Design Moderate: Sales, Admin Low: Customer Support
Continuous Learning and Development Policy	₹1,000,000	+0.95 points	9.5% reduction in attrition	High Impact: All departments Highest: Technical, Sales
Comprehensive Health and Wellness Policy	₹1,500,000	+1.0 point	10% reduction in attrition	Consistent Impact: All departments
Inclusive and Diverse Hiring Policy	₹800,000	+0.85 points	8.5% reduction in attrition	High Impact: Technical, Design Moderate: All others

5. Understanding Decision Linkages and System Dynamics

Understanding how decisions in one area impact others is crucial for successful gameplay:

1. **Demand → Supply Connection:**
   - Strategic objectives in Demand Forecasting determine the additional hours needed
   - These hours translate to specific headcount requirements in Supply Forecasting
   - Example: If Technical department needs 15,000 additional hours, and one employee works 1,760 hours (8 hours × 220 days), you'll need approximately 9 new employees

2. **Supply → Analysis & Planning Connection**:
   - The gap identified in Supply Forecasting determines how many people you need to hire
   - This directly impacts which hiring channels you should select
   - Example: For a large number of technical hires, you might use multiple channels like Job Portals and Campus Recruitment

3. **Analysis & Planning → Implementation Connection**:
   - Your hiring channels affect how quickly positions are filled
   - Salary decisions impact employee satisfaction and retention
   - Both influence which training programs will be most effective
   - Example: If hiring many new technical staff, training in advanced technologies becomes crucial

4. **Complete Cycle Impact**:
1. Training decisions affect employee efficiency, which impacts future demand
2. Compensation decisions affect attrition, which impacts future supply needs
3. HR policies influence overall satisfaction, affecting retention and productivity
4. All decisions impact costs, which must be balanced against performance

Summary on Core Decision Areas & Key Parameters

Decision Area	Key Parameters	Critical Insights
Demand Forecasting	• Organizational structure • Strategic objectives •Additional hours required • Current employee efficiency	Average employee works 8 hours/day for 220 days annually Administration requires 1 employee for every 10 new hires Must align workforce needs with business expansion goals
Supply Forecasting	• Attrition rates • Promotion rates • Internal transfers • Employee satisfaction • Time to fill positions	Identify gap between supply and demand Determine optimal new hire numbers by department Contract workers cost 12% more than permanent staff
Recruitment & Compensation	• Hiring channels • New hire salary increases • Existing employee salary increases • Budget constraints	Channel selection impacts time-to-fill and candidate quality Salary decisions affect retention and satisfaction Each recruitment channel has different cost/benefit ratio
Training & Development	• Training program selection • Program costs • Department-specific needs	Programs improve employee efficiency Must match training to strategic initiatives Low participation (<50%) indicates mismatch with needs
HR Policy Implementation	• Policy selection • Implementation costs • Employee wellbeing focus	Policies impact employee satisfaction Balance cost with long-term benefits Align with company culture

6. Performance Analysis Framework

Key Performance Indicators (KPIs):

After each round, participants should review the results in the **Report** and **Synopsis** tabs to assess the impact of their decisions. This process encourages reflection on what worked well, what didn't, and how to improve in the next round. It's not just about understanding what happened—but why it happened.

1. Employee Satisfaction Score

Description: Measures overall employee contentment influenced by compensation, training, and HR policies.
Target Range: 7.5-9.5
Strategic Impact: Higher scores improve retention and performance.

Measurement & Calculation:
- Primary contributors:
- Compensation hikes provided to existing and new employees
- Training program selection and relevance to department needs
- HR policies implemented (e.g., Remote Work, Health & Wellness)
- Tracked in the Report section under KPIs

Actions to improve:
- Match training programs to departmental needs
- Implement employee-centric policies (Health & Wellness policy has higher impact)
- Provide competitive compensation increases above industry average

Decision Connection:
- Implementation tab decisions directly influence this metric
- Balance incremental salary increases against budget constraints
- Training selection should align with strategic objectives

2. Time to Fill Positions

Description: Average days to fill vacancies across all departments.
Target Range: <39 days
Strategic Impact: Shorter times indicate effective recruitment channel selection.

Measurement & Calculation:

- Tracked in the Report section under KPIs
- Influenced by:
- Selection of hiring channels in Analysis & Planning
- Mix of recruitment methods (Online, Campus, Referral, Agency)
- Alignment with department-specific needs
- Shorter fill times reduce reliance on contract employees

Decision Connection:

- Online Job Portals typically have faster fill times but may have quality tradeoffs
- Campus Recruitment may have longer lead times but better-quality matches
- Using multiple channels simultaneously can optimize this metric
- Critical for meeting expansion goals shown in Strategic Objectives

3. Average Efficiency

Description: Productivity relative to organizational goals, measured as a percentage.
Target Range: >85%
Strategic Impact: Reflects quality of workforce planning, recruitment, and training alignment.

Measurement & Calculation:

- Tracked in the Report section under KPIs
- Calculated from:
- Match between workforce capability and strategic demands
- Department-specific efficiency levels
- Training program effectiveness
- Correct workforce sizing based on demand forecasts

Decision Connection:

- Directly tied to accuracy of demand and supply forecasting
- Training programs that match departmental needs improve this metric
- Overstaffing or understaffing negatively impacts efficiency
- References the staff efficiency charts in Demand Forecasting section

4. Training Participation

Description: Percentage of workforce engaged in training programs.
Target Range: >50%
Strategic Impact: Shows alignment of programs with employee needs and organizational goals.

Measurement & Calculation:

- Tracked in the Report section under KPIs
- Influenced by:
- Relevance of selected training to departmental needs
- Balance of technical vs. soft skills training
- Investment level in training programs
- Alignment with strategic initiatives

Decision Connection:

- Implementation tab training selection directly impacts this metric
- Consider department-specific needs from Demand Forecasting
- Strategic Objectives tab highlights technological priorities that require specific training
- Higher participation drives efficiency and satisfaction simultaneously

5. Cost Per Employee

Description: Total HR spending per employee, including salaries, training, and policies.
Target Range: Benchmark against industry standards (optimize toward lower costs while maintaining performance)
Strategic Impact: Measures cost efficiency of HR strategies.

Measurement & Calculation:

- Tracked in the Report section under KPIs
- Components include:
- Base salaries and incremental increases
- Training program costs
- Policy implementation costs
- Recruitment/branding expenses
- Contract employee costs

Decision Connection:

- Every decision area impacts this metric
- Tradeoff between cost reduction and employee satisfaction/efficiency
- Contract employees affect this differently than permanent employees
- Training investments affect short-term costs but improve long-term efficiency

6. Recruitment Effectiveness

Description: Percentage of positions filled through selected channels vs. contract staffing.
Target Range: >80%
Strategic Impact: Lower rates suggest ineffective channel selection or misaligned hiring strategy.

Measurement & Calculation:

- Tracked in the Report section under "Recruitment metrics"
- Calculated as:
- Positions successfully filled through selected channels / Total positions needed
- Higher percentages indicate better alignment of hiring channels with needs
- 100% means all positions filled through planned channels without contract reliance

Decision Connection:

- Analysis & Planning tab channel selection directly impacts this
- Higher effectiveness reduces reliance on contract staffing
- Balancing multiple channels improves overall effectiveness
- Must be evaluated alongside Time to Fill metric

Summary of Key Performance Indicators:

Metric	Description	Target Range	Strategic Impact
Employee Satisfaction Score	Measures overall employee contentment	7.5-9.5	Higher scores improve retention and performance
Time to Fill Positions	Average days to fill vacancies	<39 days	Shorter times indicate effective recruitment
Average Efficiency	Productivity relative to goals	>85%	Reflects quality of workforce planning and training
Training Participation	% of workforce in training	>50%	Shows alignment of programs with employee needs
Cost Per Employee	Total HR spending per employee	Benchmark against industry	Measures cost efficiency of HR strategies
Recruitment Effectiveness	% of positions filled through selected channels	>80%	Lower rates suggest ineffective channel selection

7. Theoretical Connections by Simulation Section

Section	Simulation Area	Theory	Summary
Business Context	Market Overview	Strategic Human Capital Theory	Human capital decisions must align with business strategy to create competitive advantage.
	Industry Analysis	Resource-Based View	Organizational success depends on developing unique, valuable, and difficult-to-imitate human resources.
	Technology Adoption	Socio-Technical Systems Theory	Effective technology implementation requires balanced attention to both technical and social systems.
Demand Forecasting	Strategic Objectives	Human Resource Planning Theory	Workforce needs should be derived from organizational strategy and translated into specific staffing requirements.
	Efficiency Analysis	Productivity Measurement Theory	Employee productivity metrics help predict future workforce requirements and identify performance gaps.
	Organizational Structure	Organizational Design Theory	Structure determines staffing needs and impacts communication, decision-making, and operational efficiency.
Supply Forecasting	Attrition Analysis	Employee Turnover Theory	Turnover patterns are predictable based on job characteristics, satisfaction levels, and labor market conditions.
	Internal Mobility	Career Development Theory	Career pathways influence employee retention and impact the organization's talent pipeline.
	Labor Market Analysis	Labor Market Economics	External talent availability affects recruitment strategy, compensation decisions, and internal development priorities
Recruitment & Compensation	Recruitment Channel Selection	Recruitment Process Theory	Explains how organizations attract, evaluate, and select candidates by focusing on fit, fairness, while emphasizing a positive candidate experience. Different channels yield varying candidate pools in terms of quality, diversity, and time-to-hire.
	Compensation Strategy	Equity Theory	Perceived fairness in compensation affects employee motivation, performance, and retention.
	Offer Competitiveness	Efficiency Wage Theory	Above-market compensation can reduce turnover, increase productivity, and attract higher-quality candidates.

Training & Development	Program Selection	Human Capital Development Theory	Strategic training investments enhance employee capabilities and organizational performance.
	Skill Gap Analysis	Competency Theory	Identifying critical competency gaps enables targeted development interventions and succession planning.
	ROI Measurement	Training Evaluation Theory	Effective training produces measurable improvements in individual performance and organizational outcomes.
HR Policy Implementation	Wellness Initiatives	Organizational Support Theory	Perceived organizational support through policies enhances commitment, engagement, and discretionary effort.
	Flexible Work Arrangements	Work-Life Balance Theory	Policies supporting work-life balance improve retention, reduce burnout, and increase attraction of talent.
	Diversity & Inclusion	Inclusion Theory	Inclusive policies enhance innovation, decision quality, and organizational adaptability.
Performance Analysis	Satisfaction Metrics	Job Satisfaction Theory	Employee satisfaction influences performance, citizenship behaviors, and turnover intentions.
	Efficiency Measurement	Productivity Analysis	Workforce productivity is influenced by skills, engagement, and alignment with organizational systems.
	Cost Management	Productivity Analysis	HR investments should be evaluated based on their returns in productivity, retention, and organizational performance.

Critical Teaching Point

- Emphasize to students that the highest-performing HR strategies demonstrate strong alignment across all decision areas, creating a self-reinforcing talent management ecosystem rather than a collection of independent tactical choices.
- When debriefing, guide participants to trace the connections between their decisions using this integrated HR framework. For example:

1. How did your business strategy analysis inform your workforce demand forecasting?
2. How did your talent supply decisions influence your recruitment channel selection?
3. How did your compensation strategy support both your recruitment effectiveness and retention goals?
4. How did your training investments align with your internal mobility targets?
5. How did your HR policy implementation complement your organizational culture and staffing approach?

8. Sample Optimal Solution

It is essential to acknowledge that numerous combinations can lead participants to achieve the desired outcomes, and the provided solution serves as a reference for faculty and is not intended to be shared directly with participants.

Please Note: The solution below is for the default Cesim Case.

Supply Forecasting	
Sales & Marketing	97
Logistics & Supply Chain	51
Design & Production	60
Technical	116
Customer Experience	40
Administration	37
Analysis & Planning	
Online Job Portals	Yes
Campus Recruitment	Yes
Employee Referral Programs	No
Social media and Company Website	No
Recruitment Agencies and Headhunters	No
Sales & Marketing, New Joinee Hike %	9
Logistics & Supply Chain, New Joinee Hike %	8
Design & Production, New Joinee Hike %	9
Technical, New Joinee Hike %	10
Customer Experience, New Joinee Hike %	8
Administration, New Joinee Hike %	7
Sales & Marketing, Existing Employee Hike %	8
Logistics & Supply Chain, Existing Employee Hike %	7
Design & Production, Existing Employee Hike %	8
Technical, Existing Employee Hike %	8
Customer Experience, Existing Employee Hike %	7
Administration, Existing Employee Hike %	6
Implementation	
Digital Marketing Mastery	Yes
Advanced E-commerce Analytics	No
Augmented Reality (AR) in Fashion E-commerce	Yes
Customer Service Excellence	Yes
Sustainable Fashion and Ethical Production	No
Remote Work Flexibility Policy	No
Continuous Learning and Development Policy	No
Comprehensive Health and Wellness Policy	Yes
Inclusive and Diverse Hiring Policy	Yes

Decision Making Rationale

For Instance, to determine the new hike percentage for the **Sales & Marketing** department based on the given data, we can approach the calculation systematically:

A. **Proposed Hike for Existing Employees:**

Key Considerations:

i. **Attrition Rate (8.20%)**: Moderate attrition indicates the need to remain competitive in compensation to retain employees.

ii. **Previous Year vs. Market Average Salary**: Current average salary for Sales & Marketing is ₹925,000, which is **3.1% higher** than the market average of ₹897,250. This suggests that the department is slightly ahead of the market, but this edge should be maintained given the attrition rate.

iii. **Company vs. Market Hike**: The **company average hike (6.3%)** is below the **market average hike (7.5%)**. To remain competitive, the hike for Sales & Marketing should lean closer to the market average, especially given the strategic importance of the department. So proposed Hike for the Existing employees

- **Baseline Hike**: Start with the company average (6.3%).
- **Market Adjustment**: Add a **1% increment** to align closer to the market hike (7.5%).
- **Retention Incentive**: Add **0.5%** to address the moderate attrition rate and ensure retention of skilled employees in this key department.
- **Proposed Hike for Existing Employees = 6.3% + 1% + 0.5% = 7.8%**

B. **Proposed Hike for New Joiners:**

Key Considerations:

i. **Market Positioning**: **Current Average Salary** for Sales & Marketing (₹925,000) is already **3.1% above the market average** (₹897,250). These existing premium positions the company competitively, even without increasing the hike percentage significantly.

ii. **Market Dynamics**: Future employees often expect compensation slightly above the **current market benchmark** to make a move. A hike above the **7.5% market hike** can be attractive but should balance cost efficiency.

iii. **Internal Parity**: Offering a significantly higher hike for future employees can lead to internal dissatisfaction and parity issues with existing staff. Ensure that any difference in hikes is justified by skills, experience, or criticality.

Proposed Adjustment: To remain competitive while attracting new talent:

- **Increase the future hike slightly above the market average**, while being mindful of the current employees' hike.
- A **0.5%–1.0% increment** over the market hike would make the offer attractive without excessively widening the gap.

Recommended Future Employee Hike:

- Market Average Hike: **7.5%**.
- Additional Competitive Edge: **+0.5% to 1.0%**.
- **Proposed Hike for Future Employees = 8.0% to 8.5%**.

9. Common Pitfalls and Teaching Interventions

Pitfall 1: Strategic-Operational Disconnect

- **Symptoms:** Students develop workforce plans disconnected from the company's technology transformation strategy.
- **Intervention:** "Map each of your HR decisions directly to a business objective. How specifically does your technical staffing plan support the AR/AI implementation? Which departments need the most significant adjustments to align with strategic priorities?"

Pitfall 2: Uniform Approach Across Departments

- **Symptoms:** Students apply identical growth percentages, compensation increases, or training programs across all departments despite different needs.
- **Intervention:** "Compare the market conditions, attrition rates, and strategic importance of your different departments. What evidence supports a differentiated approach? How might a more targeted strategy improve your overall results?"

Pitfall 3: Short-term Cost Focus Over Strategic Investment

- **Symptoms:** Students consistently choose lowest-cost options in recruitment, compensation, and training without considering long-term implications.
- **Intervention:** "Calculate the full lifecycle cost of your approach, including turnover expenses, productivity impacts, and recruitment costs. How does investing strategically in key areas now affect your three-year financial projections?"

Pitfall 4: Isolated Decision-Making

- **Symptoms:** Students make decisions in each simulation area without considering interdependencies (e.g., aggressive internal promotion without supporting training).
- **Intervention:** "Create a decision matrix showing how your choices in one area impact others. If you promote 50% of technical roles internally, what training is needed? How do your compensation decisions support your recruitment channel strategy?"

Pitfall 5: Data Misinterpretation or Underutilization

- **Symptoms:** Students overlook critical metrics like department-specific attrition rates or make decisions contradicting available data.
- **Intervention:** "Let's revisit the key data points for each department. What patterns do you see in the attrition data? How confident are you in your interpretation? What additional information would strengthen your analysis before making this decision?"

10. Instructor Tips and Insights

1. **Pre-simulation Hypothesis Exercise**: Before revealing simulation metrics, ask students to predict which HR decisions will have the greatest impact on employee satisfaction and efficiency. During debriefing, highlight disparities between predictions and actual outcomes to challenge common HR misconceptions.
2. **Real-World HR Cases**: Prepare 2-3 examples of actual companies facing similar HR challenges. Compare student strategies with these real-world approaches during debriefing.
3. **ROI Analysis Workshop**: After completing the simulation, guide students through calculating the ROI on different HR investments (recruitment channels, training programs, compensation strategies). This strengthens the business partner mindset critical for HR professionals.
4. **Policy Impact Matrix**: Have students create a matrix that maps HR policies to specific business outcomes based on their simulation results. This helps them understand the strategic impact of seemingly "soft" HR initiatives.
5. **Competitive Benchmarking**: Create a class-wide benchmarking exercise where teams compare their HR metrics (cost-per-hire, time-to-fill, satisfaction scores) against their peers. This simulates the competitive talent landscape HR professionals navigate.
6. **Strategic Alignment Check**: Before students submit final decisions, have them complete a quick alignment assessment: "How does each HR decision support the company's technological transformation?" This reinforces HR's strategic role.

Suggested Discussion Starters

1. How do market trends affect HR demand forecasting accuracy?
2. What is the impact of turnover and attrition on HR demand forecasting?
3. How can AI improve the reliability of HR demand forecasting?
4. What strategies balance internal growth and external recruitment in supply forecasting?
5. How does organizational structure aid HR in talent planning?
6. How should workforce planning adapt to shifts in business strategy?
7. How do efficiency metrics help predict workforce needs?
8. How do recruitment channels influence employer branding and image?
9. How does employee training ensure workforce agility and adaptability?
10. How does training reduce gaps between forecasted and actual workforce needs?

Frequently Asked Questions (FAQs)

General Simulation Questions

Q: How many rounds will we play in this Micro simulation?

A: The simulation typically consists of multiple rounds with maximum of 10 rounds, allowing you to refine your strategy based on previous results. Your instructor will specify the exact number of rounds for your course.

Q: How is the final score calculated?

A: Your performance is evaluated based on four key metrics: Employee Satisfaction Score, Average Time to Fill Positions, Average Efficiency Across Organization, and Cost Per Employee while the leaderboard is prepared based on the first KPI Employee Satisfaction Score. The simulation also assesses your decision-making process through ratings on Rigor, Structuring, Synthesis, and Business Judgment.

Q: Can I change my decisions after submission?

A: No, once decisions are submitted for a round, they cannot be altered. This design element emphasizes careful planning and thorough consideration before finalizing decisions.

Strategy Questions

Q: Is it better to hire through cheaper channels or more expensive ones?

A: This depends on your priorities. Cheaper channels like online job portals have lower costs per hire but may take longer to fill positions and might not attract the highest-quality candidates. Expensive channels like recruitment agencies offer faster hiring and potentially better candidates but at significantly higher costs. Most successful strategies use a mix of channels tailored to different positions.

Q: How much should I increase salaries each round?

A: This depends on multiple factors including market salary trends, current employee satisfaction, and your budget constraints. Examine the elasticity data to understand the impact of different percentage increases. Generally, staying within 3-5% of market rates maintains competitiveness while managing costs.

Q: Should I prioritize training or HR policies?

A: Both serve different purposes. Training directly improves employee efficiency and impacts specific departmental performance, while HR policies tend to have broader effects on satisfaction and retention. The optimal approach typically includes some investment in both, with training focused on departments crucial to current strategic objectives.

Technical Questions

Q: How do I calculate the exact number of employees needed in each department?
A: Start with the additional hours required (shown in Demand Forecasting) and divide by the annual working hours per employee (1,760 hours = 8 hours × 220 days). Then adjust for efficiency levels and expected attrition, promotion, and transfer rates. Remember that the Administration department requires 1 employee for every 10 new hires.

Q: What happens if I don't hire enough employees?
A: Understaffing leads to reduced organizational efficiency, potential delays in strategic initiatives, and increased workload on existing employees, which can negatively impact satisfaction. The simulation will fill critical positions with contract employees at a higher cost (12% of standard salary).

Q: How exactly does employee satisfaction impact performance?
A: Employee satisfaction affects attrition rates, efficiency levels, and the organization's ability to attract talent. Higher satisfaction scores (above 8.0) typically reduce attrition by 15-20% and improve efficiency by 5-10% compared to average satisfaction levels.

Conceptual Questions
Q: How realistic is this simulation compared to actual HR planning?
A: The simulation captures the core dynamics and trade-offs of real-world HR planning, including the relationships between recruitment, compensation, training, and employee satisfaction. While simplified for educational purposes, the causal relationships and metrics reflect those used in actual HR departments, particularly in growing companies.

Q: How do I balance short-term costs with long-term benefits?
A: This is a fundamental challenge in HR management. Generally, investments in proper recruitment channels, competitive compensation, and well-designed policies yield returns through higher retention, improved efficiency, and reduced recruitment needs. The simulation rewards strategic thinking that considers these longer-term impacts.

Q: What's the most important metric to focus on?
A: Rather than focusing on a single metric, successful strategies balance multiple objectives. However, Employee satisfaction score and Average Efficiency Across Organization often serves as a good overall indicator as it reflects the combined effects of appropriate staffing, compensation, and training decisions.

Conclusion

The **Cesim Human Resource Planning simulation** is a dynamic, experiential tool designed to enhance HR education by bridging the gap between theoretical knowledge and practical application. By placing participants in the role of HR decision-makers, the simulation creates an immersive learning environment that builds essential strategic and analytical competencies required for success in modern HR roles.

Participants develop four key competencies through the simulation:

- **Experiential Learning**: They engage in realistic HR scenarios, honing critical thinking and decision-making skills that mirror real-world business challenges.
- **Strategic Insight**: The simulation teaches how to align HR strategies with organizational goals, managing trade-offs between cost efficiency, employee satisfaction, and operational performance.
- **Systems Thinking**: It emphasizes the interconnected nature of HR functions, encouraging a holistic view of their impact on the organization.
- **Applied Analytics**: Participants use data and analytics to forecast workforce needs and evaluate the effectiveness of HR initiatives, strengthening their data literacy and analytical capabilities.

Seamlessly integrated into academic programs and enhanced through guided debriefings, the Cesim Human Resource Planning simulation prepares participants for future HR leadership. It equips them with the strategic mindset, practical experience, and analytical tools that employers expect in today's evolving workplace.